

RESEARCH STATEMENT

Hanbaek Lyu

I work in the fields of discrete probability and optimization with applications in machine learning. My research program comprises 33 journal publications—appearing in venues such as *Nature Communications*, the *Annals of Probability*, the *Annals of Applied Probability*, *Forum of Mathematics*, *Sigma*, *Transactions of the AMS*, *IMRN*, the *SIAM Journal on Optimization*, and *JMLR*—together with 15 machine-learning conference proceedings (including 8 in *ICML* and 2 in *NeurIPS*) and 9 papers currently under review. This research has been supported by three NSF grants, most recently an **NSF CAREER Award** (DMS-2541878, \$450,000, 2026–2031) on random matrices, Schrödinger bridges, optimal transport, and generative modeling, together with industry funding from KRAFTON AI on Schrödinger-bridge-based generative modeling and online linear attention for LLMs with huge context lengths.

1. *Discrete Probability*: EMERGENCE IN LARGE SCALE SYSTEMS

In my research in discrete probability, I establish scaling limits, phase transitions, and universality classes for large discrete systems. A recurring theme in my work is establishing unexpected connections between seemingly distant mathematical objects. For instance, synchronization of coupled oscillators connects to clustering of an annihilating particle system [ALS24], and its long-term asymptotic behavior is given by the persistence probabilities of a random walk with correlated increments [LS19]. Three-color excitable media can be viewed as a projection of the classical clock synchronization algorithm due to Lamport. A two-type diffusion-limited annihilating system can be analyzed via the embedded token-passing and path-swapping processes [JLS23]. Longest soliton lengths in the randomized box-ball system are connected to the extreme behavior of a circular exclusion process and its diffusion limit as semimartingale reflecting Brownian motion [LLP22, LLPS24]. Random matrices with given margins concentrate around static Schrödinger Bridges.

1.1. Random Matrices with Given Margins. Large random matrices with fixed row and column sums (margins) arise ubiquitously in diverse contexts such as contingency tables, network flow, and graphs with specified degree sequences. These matrices can exhibit striking phase transition phenomena not seen in classical models. For instance, as conjectured by Barvinok in 2010, a small change in the margin of a random contingency table can trigger a sharp phase transition, a non-analytic behavior that suggests deep structural changes and computational hardness. With Dittmer and Pak, I have settled this conjecture in a 2020 publication in *Transactions of the AMS* [DLP20]. With Pak, I have also shown that Good’s 1950 independence heuristic for counting contingency tables fails dramatically after the phase transition [LP22]. With Mukherjee, I have discovered that there is a deeper principle governing the structure of margin-conditioned random matrices. Our key innovation is the “transference principle,” which allows us to approximate the margin-conditioned matrix using a tractable matrix with independent entries drawn from an exponential tilt of the base measure. The surprising connection I discovered was that such a required tilt is given by two Schrödinger potentials in an implicit static Schrödinger bridge/entropic optimal transport problem. Our unified framework brings techniques from the this literature, such as Kantorovich duality, Schrödinger potentials, and Sinkhorn algorithms, to solve previously intractable problems involving random matrices with given margins. This work [LM25] is currently under review in *Probability Theory and Related Fields*.

1.2. Integrable Systems and Solitons. Box-Ball Systems (BBS) are discrete integrable systems that originate from the celebrated Korteweg-de Vries (KdV) equation, crystalizing the classical soliton solutions in the integrable PDE literature in combinatorial settings. I investigate how solitons emerge and evolve in such systems from a disordered random initial state. My work with Levine and Pike [LLP22] (published in *IMRN*) is considered a foundational paper in the literature of randomized BBS, establishing the scaling limit of the soliton lengths in the 1990 Takahashi-Satsuma BBS with i.i.d. initial configurations. Recently, in a publication in *Forum of Mathematics, Sigma* (in collaboration with Lewis, Pylyavskyy, and Sen), I established the intricate phase diagram and extremely precise asymptotics for the top soliton lengths in multicolor BBS [LLPS24]. The core analysis focuses on the “multicolor carrier process,” a high-dimensional Markov chain in the integer orthant that behaves like a complex queuing system. By establishing its scaling limit and ruin probabilities, I obtained extremely precise asymptotics for the soliton lengths down to the $\log \log n$ term, a

level of precision out of reach of the current state-of-the-art Thermodynamic Bethe Ansatz. Furthermore, in [KKLL25], for the first time, I established a concrete connection between the two major universality classes of integrable probability—KdV and KPZ—by constructing a stochastic BBS that interpolates between the deterministic BBS and the PushTASEP particle system, proving convergence to a semimartingale reflecting Brownian motion in the diffusive limit.

2. Optimization and Machine Learning: ALGORITHMS TO THEORY TO APPLICATIONS

My research in optimization and machine learning revolves around matrix factorization and its conceptual generalization to nonconvex constrained optimization. For these problems, I develop various principled optimization algorithms with rigorous theory on their convergence properties. This includes online matrix/tensor factorization for Markovian data [LNB20, LSN22], supervised matrix factorization [LLY23, LLY24], stochastic first-order methods for correlated data samples [Lyu24, AL23, PL24], and block majorization-minimization [LL25] and its Riemannian extensions [LBNL23, LBNL24]. My research also bridges the gap between theory and practice, theoretically explaining why practical learning algorithms work. This includes cyclic block optimization for nonconvex problems [LL25], policy optimization in reinforcement learning [Pow25, Pow26], and coordinate ascent maximum likelihood for phylogenetic inference [CJLRS26, CJLRS25, CLR25].

2.1. Cyclic Block Optimization. Many high-impact large-scale optimization problems are often solved by some form of cyclic block optimization, where one simply optimizes for one block of parameters and then proceeds to the next block cyclically. Despite its conceptual simplicity, vast applicability, and practically proven efficacy, its theoretical understanding has surprisingly been limited compared to the random coordinate methods pioneered by Nesterov, Polyak, and Wright. With my student Li, I have developed a unique analysis technique that has settled the theoretical speed of convergence for this type of algorithm for the most challenging and practical setting of nonconvex nonsmooth constrained problems. This work is published in SIAM Journal on Optimization [LL25], one of the top two journals in optimization. With Li, I have further extended these results to Riemannian manifolds, a mathematical setting that captures intricate geometry in the parameter space [LBNL23, LBNL24] (published/accepted in top machine learning venues of ICML and JMLR). This theory provided the first iteration complexity bound for the Bures-JKO scheme for Wasserstein variational inference with non-strongly convex potentials.

2.2. Stochastic Optimization. For large-scale data streams where data arrives sequentially in real-time, one needs to develop stochastic learning algorithms that process the data sequentially and gradually learn the best parameter that explains the overall stream. A major challenge in this domain is that real-world data streams (e.g., web crawling or network sampling) often violate the standard i.i.d. assumptions used in optimization theory. In a series of JMLR publications [LNB20, LSN22, Lyu24], I developed a general family of stochastic block majorization-minimization algorithms and proved strong theoretical guarantees for their convergence properties under non-i.i.d. data settings. This unifying framework covers a wide range of classical and modern algorithms such as Nesterov’s double-averaging, proximal point empirical loss minimization, online expectation-maximization, and online matrix and tensor factorization.

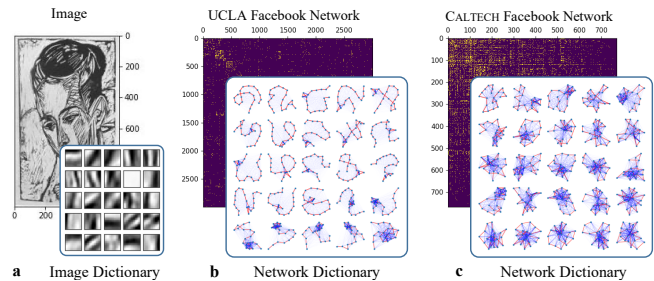


Figure 1. Illustration of learned dictionary from square patches of an image and 20-node subgraphs of networks.

2.3. High-impact Machine-Learning Applications. Based on my research in discrete probability and non-convex optimization, I also develop high-impact machine learning applications. For instance, I have developed innovative methods for decomposing networks into a small set of key connection patterns and

reconstructing the networks from them. A culmination of this research was published in *Nature Communications* (featured in the Editor’s Highlight) [LKVP24], where a reviewer praised it as a “tour de force” in the field. The core idea is to sample a large number of subgraph patterns using random k -paths [LMS23] and then extract a small number of essential patterns using an online matrix factorization algorithm [LNB20]. The extracted patterns reveal how the network is composed at the mesoscale. A specific application of this method is learning single-molecule chromatin interaction patterns [PPL⁺]. Also, using supervised matrix factorization, I have developed methods for identifying latent gene groups that are highly correlated with a specific type of cancer [LLY23, LLY24] (in NeurIPS and ICML), which turns out to identify some clinically known oncogenes.

3. RESEARCH PROGRAM

My research over the next five years is organized around three directions. The first is the program of my NSF CAREER Award on random matrices and Schrödinger bridges; the second carries my work on box-ball systems toward the central questions of integrable probability; the third is an industry collaboration with KRAFTON AI on the architecture of large language models.

3.1. Random Matrices, Schrödinger Bridges, and Entropic Optimal Transport. My CAREER program (DMS-2541878, 2026–2031) rests on a single observation: margin-constrained random matrices, static Schrödinger bridges, entropic optimal transport, and generative modeling are four views of one object. The transference principle of [LM25] identified the typical structure of a margin-conditioned random matrix with the Schrödinger potentials of an implicit bridge. With my students Duan and Powell, I have since built a limit theory on both sides of this correspondence. For Sinkhorn-rescaled random matrices with independent entries, we proved concentration around the rescaled mean with explicit non-asymptotic rates, convergence to the continuous static Schrödinger bridge, bulk rigidity for the associated sample covariance matrix, and a central limit theorem for the empirical Schrödinger potentials [DLP26b]. The engine is a new quantitative stability theory for the bridge itself—Lipschitz in the Hellinger distance with respect to the reference measure, Hölder-1/2 with respect to the margins, and stable in L^∞ at the level of the discrete potentials—which is of independent interest in entropic optimal transport.

For conditioned ensembles, we studied Poisson random matrices conditioned on their margins [DLP26a]. Through Barvinok’s permanent identity, such a table is the block-count matrix of a random permutation drawn from a Gibbs measure of Mallows type, and this correspondence allows us to analyze the table at three scales: a transference principle to independent Poisson entries whose means form the typical table; a quantitative law of large numbers toward a static Schrödinger bridge; and a central limit theorem jointly for all bounded test functions, whose limiting field is white noise with intensity given by the bridge, conditioned to have vanishing marginals. The key ingredient is an asymptotic formula for the partition function as a Fredholm determinant of the canonical correlations of the bridge. For unit margins, it yields Pal’s conjecture on the limiting partition function of Mallows-type random permutations [Pal24] under weaker assumptions, together with a central limit theorem for their linear statistics.

On the algorithmic side of the program, with my student Bi and with Jakwang Kim, I am developing Sinkhorn-type methods for entropic optimal transport with *Lagrangian costs*—costs defined as the least action of a path, such as shortest-path distances on large graphs, geodesic distances on data manifolds, and optimal control costs—for which each evaluation of the cost is itself an optimization problem, so that the Gibbs kernel cannot even be formed [BKL26]. Our *stochastic mesoscale approximation* replaces the Kantorovich dual by a stochastic surrogate whose Gibbs kernel is the expectation of cheap local kernels on randomly sampled blocks. The surrogate is itself an entropic transport problem with the same marginals and a perturbed kernel, so its accuracy reduces to the kernel stability of the Schrödinger bridge—the same question that drives the random-matrix theory above—and we show that the error is controlled by the expected cost bias seen by the true bridge. We solve the surrogate by *stochastic local Sinkhorn* algorithms, which solve one sampled block per iteration and patch the result into the current potentials: at a fixed step

size the law of the iterates equilibrates geometrically, and with diminishing step sizes the iterates converge at the Monte Carlo rate. Next, I aim to extend these certificates from surrogates that only overestimate the cost to two-sided kernel perturbations, such as Nyström and sparsified kernels, which would certify widely used fast Sinkhorn variants at the level of the transport plan and account for the bias of minibatch optimal transport.

Over the next five years I will pursue three questions that these results bring within reach. The first is the *sparse regime*. When the average margins stay bounded—as for random permutations and degree-constrained bipartite graphs—the transference exponent becomes of the same order as the dimension, and the table becomes a sparse integer matrix whose spectrum is no longer governed by a Dyson equation. There the permutation representation should take the place of transference, and I expect the limiting spectrum to be that of the local weak limit of a weighted configuration model; proving this requires concentration and local weak convergence for the underlying Gibbs measure on permutations. The second is the *local fluctuations of conditioned ensembles*. Transference transfers only exponentially rare events, so the local laws and rigidity estimates available for independent entries, which hold with polynomially small failure probabilities, do not pass to the conditioned model, and our spectral results for conditioned tables currently stop at the global law. Local laws proved directly for margin-conditioned matrices would place their fluctuation theory on the same footing as classical random matrix theory. The third is *dependent ensembles*. In generative modeling and in attention mechanisms (Section 3.3), the matrix being Sinkhorn-rescaled is a kernel matrix $(k(X_i, X_j))_{i,j}$ whose rows and columns are dependent through shared sample points. Extending the stability and fluctuation theory to such ensembles connects the probabilistic core of the program to its applied endpoint: quantifying the finite-sample error of Schrödinger-bridge-based estimators and samplers.

3.2. Box-Ball Systems and Integrable Probability. Outside the CAREER program, I am pushing my work on box-ball systems toward the central questions of integrable probability. For the deterministic box-ball system with random initial data, the fluctuation theory of solitons is now well developed: Olla, Sasada, and Suda showed that a tagged soliton converges to Brownian motion in the diffusive scale and satisfies a large deviation principle [OSS24]. My program asks what happens when a small amount of stochasticity is introduced into the dynamics itself. I am developing the fluctuation theory of such stochastic box-ball systems from two perspectives: invariance principles, and the algebraic methods of KPZ fluctuation theory. With Keating, Kim, and Loeser [KKLL25], I established a general invariance principle for d -dimensional Markov jump processes converging to semimartingale reflecting Brownian motions in convex polyhedra, which accommodates distinct boundary dynamics on every face intersection and hence an overdetermined Skorokhod decomposition with more reflection vectors than faces. Applied to the stochastic box-ball system, a probabilistic interpolation between the Takahashi–Satsuma system and PushTASEP, it yields a diffusive limit whose reflection matrix records a “solitonic bias” absent from PushTASEP, together with a unified diffusion limit for exclusion processes with asymmetric collisions. With Aguirre, Kim, and Paguyo [AKLP26], I examined the second perspective. Box-ball systems with a stochastic carrier, which fails to pick up or drop off balls with prescribed probabilities, form a family of stochastic higher-spin vertex models, and the coordinate Bethe ansatz is the entry point to the algebraic methods behind known KPZ fluctuation scaling results. For the coordinate Bethe ansatz, the answer is negative. We classified the stochastic weights for which the coordinate Bethe ansatz closes at every particle number: they are exactly the fusion weights of Corwin and Petrov, a subfamily that stays a uniformly positive distance from the box-ball point at every capacity. In particular, the coordinate Bethe ansatz does not close for the stochastic box-ball system at any capacity $J \geq 2$, the regime in which solitons exist: plane-wave solvability fails strictly before the crystal limit, at which Yang–Baxter integrability survives only in combinatorial form.

These results set the agenda for the next five years. The invariance principle applies to a wide range of particle systems but does not use the integrable structure of the box-ball system, while the coordinate Bethe ansatz uses that structure but does not survive a stochastic carrier. My aim is a fluctuation theory

between the two, along three lines. First, I will extend the invariance principle to a number of particles that grows with the diffusive scaling, where the limit is an infinite-dimensional reflected diffusion, and determine whether these limits carry a random-matrix structure, as Warren’s reflected interlacing Brownian motions do for the GUE corners process and, by the work of Gorin and Shkolnikov, for multilevel TASEP and the edge of multilevel Dyson Brownian motion [War07, GS15, GS14]. Second, the algebraic route remains open beyond plane-wave closure: our analysis produced exact formulas for these vertex models, including exact recursions for the generating function at capacities $J = 2$ and 3 and a closed form for the top Bethe amplitude at every capacity, and I intend to build on them an alternative asymptotic analysis adapted to the near-box-ball corner of the lattice models. Third, the stochasticity injected in current models may be too severe to preserve the combinatorial integrability of the box-ball system; I aim to identify stochastic perturbations mild enough to retain it, so that their fluctuations can be analyzed with the combinatorial tools of the deterministic system.

3.3. Machine Learning with KRAFTON AI. Through an industry collaboration with KRAFTON AI, I work on the architecture of large language models, where tools from the first direction find new applications. The first project addresses long-context language modeling with linear attention, which replaces softmax attention by a recurrence on a fixed-size state and thereby trades the growing key–value cache for memory collisions. We introduced Integral DeltaNet, which pairs a fast delta-rule state holding current key–value associations with a slow integral state retaining a weighted history of it, combines the two adaptively at readout, and admits a chunkwise-parallel formulation for hardware-efficient training. At 340M and 1.3B parameters, trained on 15B and 100B tokens, it improves on the Kimi Delta Attention and Gated DeltaNet-2 baselines in language modeling, commonsense reasoning, and retrieval.

The second project, with my student Moon, studies Sinkhorn attention, in which softmax normalization is replaced by a doubly stochastic Sinkhorn rescaling, so that each attention matrix is a static Schrödinger bridge between uniform marginals. For i.i.d. tokens, we show that the Sinkhorn attention computed from finitely many tokens is close, with high probability, to the population-level scaling evaluated at the sampled tokens, using the potential-stability theory of [DLP26b], and we are removing the boundedness assumption on the token embeddings. The longer-term aim is a random-matrix theory of attention: fluctuation and spectral theory for Sinkhorn-rescaled kernel matrices, and a principled account of what fixed-size linear-attention states can and cannot retain.

These directions admit entry at several levels: the sparse-regime and box-ball questions are concrete enough for a second-year student, while the local-law and dependent-ensemble questions call for a postdoc or an established collaborator. I have supervised five doctoral students, two postdoctoral scholars, and twenty-four undergraduate researchers across five REU cohorts, and I structure groups so that postdocs mentor graduate students and graduate students mentor undergraduates.

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